Split-Prime Image Defects for Polynomial Image Sets

Exact criteria, finite-field classification, and the higher-degree obstruction

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Scope and conclusion

Let

$$P(x,y) = \sum_{j=0}^{m} P_j(x,y) \in \mathbb{Z}[x,y],$$

where P_j is homogeneous of degree j and $P_m \neq 0$.

The split-prime method generalizes exactly through an explicit gain-versus-height inequality. It gives a complete result for quadratic polynomials whose top form is a square, but a fixed finite-field defect $\eta < 1$ alone is not enough in degree $m > 2$.

1. Exact split-prime criterion in arbitrary degree

For $0 \le j \le m$, define the real oscillation

$$\omega_j := \max_{|u|,|v|\le 1} P_j(u,v) - \min_{|u|,|v|\le 1} P_j(u,v).$$

Thus $\omega_0 = 0$, because P_0 is constant.

Arithmetic hypothesis

Let S be a finite set of distinct rational primes. Suppose there is a sequence of totally real fields K_n such that

$$d_n = [K_n : \mathbb{Q}] \longrightarrow \infty,$$

every $q \in S$ splits completely in every K_n, and

$$\Delta_{K_n}^{1/d_n} \le D$$

for a constant D independent of n.

Put

$$Q = \prod_{q\in S} q, \qquad I_q = P_m(\mathbb{F}_q, \mathbb{F}_q), \qquad \alpha_q = \frac{|I_q|}{q}, \qquad \rho = \prod_{q\in S} \alpha_q.$$

For $T \geq 1$, set

$$W_{P,Q}(T) = \sum_{j=1}^{m} \omega_j Q^{m-j} T^j$$

and

$$\boxed{\Theta_{P,S}(T) = D\rho \, \frac{Q + W_{P,Q}(T)}{T^2}.}$$

Theorem 1: exact sufficient condition

If there exists a fixed $T > \sqrt{D}$ such that

$$\boxed{\Theta_{P,S}(T) < 1,}$$

then there exist $c_P > 0$ and arbitrarily large finite $A \subset \mathbb{R}$ satisfying

$$|P(A, A)| \leq |A|^{2-c_P}.$$

One may take

$$\boxed{c_P = \min\left\{1, \, \frac{-\log \Theta_{P,S}(T)}{\log(2T+1)}\right\} > 0.}$$

This is the strongest clean criterion delivered by the present scaling-and-residue method.

2. Proof of Theorem 1

Fix one of the fields $K = K_n$, write $d = [K : \mathbb{Q}]$, and let

$$\Phi : K \longrightarrow \mathbb{R}^d$$

be its full Minkowski embedding.

Define the centered algebraic-integer box

$$B_K(T) = \{b \in \mathcal{O}_K : |\sigma_i(b)| \leq T \text{ for all } 1 \leq i \leq d\}.$$

The use of a centered box is important: for a general polynomial, an arbitrarily translated cube could have polynomial oscillation depending on the translation.

Choose any real embedding $\sigma : K \hookrightarrow \mathbb{R}$ and set

$$\boxed{A_K = \left\{\frac{\sigma(b)}{Q} : b \in B_K(T)\right\}.}$$

2.1 Cardinality of A_K

The lattice $\Phi(\mathcal{O}_K)$ has covolume $\sqrt{\Delta_K}$.

Apply Blichfeldt's averaging lemma to a cube of side T. Some translate contains at least

$$\frac{T^d}{\sqrt{\Delta_K}}$$

lattice points. Subtracting one of these points from all the others produces the same number of distinct lattice elements inside $[-T, T]^d$. Therefore

$$|B_K(T)| \geq \frac{T^d}{\sqrt{\Delta_K}} \geq \left(\frac{T}{\sqrt{D}}\right)^d.$$

Distinct algebraic integers are 1-separated in L^∞, since

$$1 \le |N_{K/\mathbb{Q}}(b-b')| = \prod_{i=1}^{d} |\sigma_i(b-b')|.$$

Packing unit cubes gives

$$|B_K(T)| \le (2T+1)^d.$$

Because σ and division by Q are injective,

$$\boxed{\left(\frac{T}{\sqrt{D}}\right)^d \le |A_K| \le (2T+1)^d.}$$

As $T > \sqrt{D}$ and $d \to \infty$, these sets are arbitrarily large.

2.2 Clearing the denominator

For $b, c \in B_K(T)$, define

$$F_Q(b,c) := Q^m P(b/Q, c/Q).$$

By homogeneity,

$$F_Q(b,c) = \sum_{j=0}^{m} Q^{m-j} P_j(b,c) \in \mathcal{O}_K.$$

For every $q \mid Q$,

$$\boxed{F_Q(b,c) \equiv P_m(b,c) \pmod{q\mathcal{O}_K}.}$$

Moreover,

$$P\left(\frac{\sigma(b)}{Q}, \frac{\sigma(c)}{Q}\right) = \frac{\sigma(F_Q(b,c))}{Q^m}.$$

The map $z \mapsto \sigma(z)/Q^m$ is injective, so

$$|P(A_K, A_K)| = |\{F_Q(b,c) : b, c \in B_K(T)\}|.$$

2.3 Archimedean output container

For each real embedding σ_i,

$$|\sigma_i(b)|, |\sigma_i(c)| \le T.$$

Since P_j is homogeneous,

$$P_j(\sigma_i(b), \sigma_i(c))$$

ranges in an interval of length at most

$$\omega_j T^j.$$

Thus

$$Q^{m-j} P_j(\sigma_i(b), \sigma_i(c))$$

ranges in an interval of length at most

$$\omega_j Q^{m-j} T^j.$$

The constant term contributes only a translation. Consequently, every coordinate of $\Phi(F_Q(b,c))$ lies in an interval of length at most

$$W_{P,Q}(T) = \sum_{j=1}^{m} \omega_j Q^{m-j} T^j.$$

2.4 Number of permitted residue classes

Because q splits completely,

$$\mathcal{O}_K/q\mathcal{O}_K \cong \mathbb{F}_q^d.$$

In every one of these d coordinates, the reduction of $F_Q(b,c)$ belongs to

$$I_q = P_m(\mathbb{F}_q, \mathbb{F}_q).$$

Therefore the number of permitted classes modulo $q\mathcal{O}_K$ is

$$|I_q|^d.$$

The Chinese remainder theorem gives

$$\#\{\text{permitted classes modulo } Q\mathcal{O}_K\} = \prod_{q\in S} |I_q|^d = (Q\rho)^d.$$

2.5 Packing within a residue class

Two distinct algebraic integers in the same class modulo $Q\mathcal{O}_K$ differ by $Q\beta$, where $0 \neq \beta \in \mathcal{O}_K$. Hence

$$\|\Phi(Q\beta)\|_\infty \geq Q.$$

Thus the points in one residue class are Q-separated.

A product of intervals of length $W_{P,Q}(T)$ contains at most

$$\left(1 + \frac{W_{P,Q}(T)}{Q}\right)^d$$

such points. Summing over the disjoint permitted residue classes,

$$\begin{aligned} |P(A_K, A_K)| &\leq (Q\rho)^d \left(1 + \frac{W_{P,Q}(T)}{Q}\right)^d \\ &= \left[\rho(Q + W_{P,Q}(T))\right]^d. \end{aligned}$$

Since

$$|A_K|^2 \geq \frac{T^{2d}}{D^d},$$

we obtain

$$\boxed{|P(A_K, A_K)| \leq \Theta_{P,S}(T)^d |A_K|^2.}$$

Finally,

$$|A_K| \leq (2T+1)^d \quad \Longrightarrow \quad d \geq \frac{\log|A_K|}{\log(2T+1)}.$$

If $\Theta_{P,S}(T) < 1$, this yields

$$|P(A_K, A_K)| \leq |A_K|^{2-c_P}.$$

3. Uniform finite-field defect

Suppose

$$|P_m(\mathbb{F}_q, \mathbb{F}_q)| \leq \eta q$$

for every $q \in S$, where $0 < \eta < 1$. If $t = |S|$, then

$$\rho \leq \eta^t.$$

The exact sufficient inequality becomes

$$\boxed{D\eta^t \left(\frac{Q}{T^2} + \sum_{j=1}^{m} \omega_j Q^{m-j} T^{j-2}\right) < 1.}$$

Equivalently,

$$t\log(1/\eta) > \log D + \log\left(\frac{Q + W_{P,Q}(T)}{T^2}\right).$$

This is the correct general prime-count versus root-discriminant inequality. Notice that for $m > 2$, the right side also depends on the sizes and product of the selected primes through Q. The number t alone is not sufficient.

4. The quadratic case

Suppose $m = 2$. Then

$$W_{P,Q}(T) = \omega_2 T^2 + \omega_1 QT,$$

and hence

$$\Theta_{P,S}(T) = D\rho\left(\omega_2 + \frac{\omega_1 Q}{T} + \frac{Q}{T^2}\right).$$

For a fixed set S, we may let T be as large as needed. Therefore the precise asymptotic condition is

$$\boxed{D\rho\,\omega_2 < 1.}$$

Under the uniform estimate $\alpha_q \le \eta$, it is sufficient that

$$\boxed{D\omega_2\eta^t < 1.}$$

If D is fixed independently of t, the required number of primes is

$$\boxed{t > \frac{\log(D\omega_2)}{\log(1/\eta)}.}$$

More precisely, when $D\omega_2 \ge 1$, one may take

$$t_{\min} = \left\lfloor \frac{\log(D\omega_2)}{\log(1/\eta)} \right\rfloor + 1.$$

If the tower depends on an auxiliary parameter ℓ, the correct condition is

$$t_\ell \log(1/\eta) > \log D_\ell + \log\omega_2.$$

In the supplied field construction,

$$t_\ell \asymp \ell^2, \qquad \log D_\ell = O(\ell\log\ell).$$

Thus, for every fixed $\eta < 1$,

$$t_\ell \log(1/\eta) - \log D_\ell \longrightarrow +\infty.$$

So the quadratic condition is eventually satisfied.

5. Classification of homogeneous forms with persistent image defect

Let $F(x, y) \in \mathbb{Z}[x, y]$ be a nonzero homogeneous binary form of degree m.

Factor it over $\overline{\mathbb{Q}}$ as

$$F(x, y) = c\prod_{\nu=1}^{s} L_\nu(x, y)^{e_\nu},$$

where the L_ν are pairwise nonproportional linear forms. Define

$$r(F) := \gcd(e_1, \dots, e_s).$$

Then $r(F) > 1$ exactly when F is a proper power:

$$F = cG^r$$

for some $r > 1$ and some homogeneous G, after absorbing a scalar into c.

Theorem 2: finite-field image dichotomy

The following are equivalent:

1. There exist $\eta < 1$ and infinitely many primes q such that

$$|F(\mathbb{F}_q, \mathbb{F}_q)| \leq \eta q.$$

2. F is a proper power over $\overline{\mathbb{Q}}$.
3. $r(F) > 1$.

More precisely:

- If $r(F) = r > 1$, then for every sufficiently large prime $q \equiv 1 \pmod{r}$,

$$|F(\mathbb{F}_q, \mathbb{F}_q)| \leq 1 + \frac{q-1}{r}.$$

- If $r(F) = 1$, then

$$\boxed{F(\mathbb{F}_q, \mathbb{F}_q) = \mathbb{F}_q}$$

  for every sufficiently large prime q.

Proof: the proper-power direction

If

$$F = cG^r$$

and $q \equiv 1 \pmod{r}$, every nonzero value of F belongs to the coset

$$c(\mathbb{F}_q^\times)^r.$$

Hence

$$|F(\mathbb{F}_q^2)| \leq 1 + \frac{q-1}{r}.$$

For example, one may take

$$\eta = \frac{r+1}{2r} < 1.$$

Proof: the non-power direction

Let

$$g = \gcd(m, q-1), \qquad H = (\mathbb{F}_q^\times)^m.$$

The subgroup H has index g.

For any projective direction $[x : y]$ with $F(x, y) \neq 0$, scaling by $\lambda \in \mathbb{F}_q^\times$ gives

$$F(\lambda x, \lambda y) = \lambda^m F(x, y).$$

Thus all values in the coset

$$F(x, y)H$$

occur.

It remains to show that every coset of H is represented by some projective direction.

We use the following standard Weil character-sum result:

If χ is a nontrivial multiplicative character of order r and $f(t) \in \mathbb{F}_q(t)$ is not an r-th power over $\overline{\mathbb{F}_q}(t)$, then

$$\left| \sum_{t \in \mathbb{F}_q} \chi(f(t)) \right| \leq C_{\deg f} \sqrt{q},$$

where the constant depends only on the degree and factorization complexity of f.

Characters of $\mathbb{F}_q^\times$ that are trivial on H have order dividing g, hence dividing m. If F is not a proper power, then for all sufficiently large primes its reduction is not an r-th power for any $r > 1$ dividing m.

Character orthogonality therefore shows that the number N_C of projective directions whose nonzero F-value belongs to a specified coset C of H satisfies

$$N_C = \frac{q + O_m(\sqrt{q})}{g}.$$

Since $g \leq m$, this is positive for all sufficiently large q. Hence every coset of H occurs, so every nonzero field element occurs. The value 0 is obtained at $(0, 0)$.

This proves eventual surjectivity.

6. Concrete classes of leading forms

Leading form P_m	Persistent defect?	Reason
$cL(x,y)^m$	Yes	Values lie in one coset of m-th powers.
$cG(x,y)^r$, $r > 1$	Yes	Values lie in a coset of r-th powers.
$x^a y^{m-a}$	Exactly when $\gcd(a, m-a) > 1$	It is a proper power exactly in this case.
A squarefree binary form	No	All linear-factor multiplicities equal 1.
$x^m + y^m$	No, for large primes	It is squarefree over characteristic 0.
xy	No	It is not a proper power; in fact it is visibly surjective.
A nondegenerate binary quadratic form	No, for large primes	It is not the square of a linear form.
$cL(x,y)^2$	Yes	Image is a scalar multiple of the squares.

Small primes at which coefficients vanish or distinct factors collide can behave exceptionally. There are only finitely many such primes, and they must be inserted individually into the exact criterion of Theorem 1.

7. Unconditional quadratic generalization

Combining Theorems 1 and 2 with the supplied field-tower construction gives the following.

Corollary. Let $P \in \mathbb{Z}[x, y]$ have degree 2, and suppose

$$P_2(x, y) = cL(x, y)^2$$

for some nonzero rational linear form L. Then there exist $c_P > 0$ and arbitrarily large finite $A \subset \mathbb{R}$ such that

$$|P(A, A)| \leq |A|^{2 - c_P}.$$

Indeed, outside finitely many bad primes,

$$|P_2(\mathbb{F}_q, \mathbb{F}_q)| = \frac{q+1}{2}.$$

The Chebotarev step in the supplied tower construction can select the split primes among primes satisfying any fixed additional cyclotomic splitting condition. In particular, it can select odd good primes. For the primes $q \equiv 1 \pmod 4$ already used there,

$$\alpha_q = \frac{q+1}{2q} \le \frac{3}{5}.$$

Since $t_\ell \asymp \ell^2$ while $\log D_\ell = O(\ell \log \ell)$,

$$D_\ell \omega_2 \left(\frac{3}{5}\right)^{t_\ell} < 1$$

for sufficiently large ℓ.

This includes

$$P_2(x,y) = (x-y)^2.$$

Linear and constant polynomials are easier: for $m = 1$, the exact criterion works even with $\rho = 1$ by taking T large, and for $m = 0$ the image has cardinality 1.

8. Why a fixed defect is insufficient for $m > 2$

The first problem is the archimedean degree mismatch.

Even in the most favorable homogeneous case $P = P_m$,

$$W_{P,Q}(T) = \omega_m T^m,$$

so

$$\Theta_{P,S}(T) = D\rho\left(\frac{Q}{T^2} + \omega_m T^{m-2}\right).$$

For $m > 2$, minimizing over $T > 0$ gives

$$T^m = \frac{2Q}{(m-2)\omega_m}$$

and

$$\min_T \left(\frac{Q}{T^2} + \omega_m T^{m-2}\right) = C_m \omega_m^{2/m} Q^{(m-2)/m},$$

where

$$C_m = \frac{m}{2}\left(\frac{2}{m-2}\right)^{(m-2)/m}.$$

Thus the optimized condition is at least

$$\boxed{DC_m \omega_m^{2/m} \prod_{q \in S} \left(\alpha_q q^{(m-2)/m}\right) < 1.}$$

But every nonzero homogeneous degree-m form satisfies, for all sufficiently large q,

$$|P_m(\mathbb{F}_q^2)| \ge 1 + \frac{q-1}{\gcd(m, q-1)} \ge 1 + \frac{q-1}{m}.$$

Indeed, choose one direction on which $P_m \neq 0$ and scale along that line.

Therefore

$$\alpha_q > \frac{1}{m},$$

and

$$\alpha_q q^{(m-2)/m} \ge \frac{q^{(m-2)/m}}{m}.$$

For sufficiently large q, this factor exceeds 1. Consequently, adding more and more large good primes eventually worsens the inequality rather than improving it.

Lower-degree terms can only add further nonnegative terms to $W_{P,Q}(T)$. In particular, if both $P_1 \neq 0$ and $P_m \neq 0$, balancing their contributions typically forces $T \asymp Q$ and creates the still larger cost Q^{m-2}.

Hence

$$\boxed{\text{For } m > 2, \quad |P_m(\mathbb{F}_q^2)| \leq \eta q \text{ with fixed } \eta < 1 \text{ is not by itself sufficient.}}$$

Some special higher-degree polynomials and small-prime configurations may satisfy the exact criterion of Theorem 1, but they must be checked using the full quantity $\Theta_{P,S}(T)$.

9. Exact boundary of the present argument

The method proves:

- all constant and linear polynomials;
- all quadratic polynomials whose leading quadratic form is a scalar multiple of a square of a linear form;
- any higher-degree polynomial satisfying the explicit split-tower inequality

$$\inf_{T>\sqrt{D}} D\rho \frac{Q + W_{P,Q}(T)}{T^2} < 1.$$

It does **not** prove the result for arbitrary polynomials.

There are two successive obstructions:

1. **Generic finite-field surjectivity.** If P_m is not a proper power, then

$$P_m(\mathbb{F}_q^2) = \mathbb{F}_q$$

for every sufficiently large prime q. Thus there is no local image defect to amplify.

2. **Degree mismatch.** Even when P_m is a proper power, its image density is only a fixed constant such as $1/r$. For $m > 2$, this generally cannot compensate for the archimedean T^m output scale relative to the T^2 input-pair scale.

The first obstruction is the earliest failure when moving toward arbitrary P; the second shows that proper-power leading forms alone do not settle higher degree.